



SOCKET

NETWORK

- A **network** is a group of interconnected devices or systems that communicate and share resources with each other.
- A **network** = **devices** + **connections** + **communication**.
- **Real-world example:** If your **mobile phone**, **laptop**, and **Wi-Fi router** are connected and share the internet, they form a **network**.

Feature	Description
Purpose	Communication and resource sharing
Basic Components	Nodes (computers/devices) and Links (cables/wireless connections)
Communication	Through signals, protocols, and network devices
Examples	Internet, office network, home Wi-Fi

TYPES OF NETWORKS

Type	Full Form / Meaning	Range
PAN	Personal Area Network	Around 1–10 meters (e.g., Bluetooth)
LAN	Local Area Network	Small areas like schools/offices
MAN	Metropolitan Area Network	City-level coverage
WAN	Wide Area Network	Country or worldwide (e.g., Internet)

IP ADDRESS

- An **IP address (Internet Protocol address)** is a **unique numerical identifier assigned to each device connected to a network**.
- It allows devices to **locate, identify, and communicate** with each other over the internet or any computer network.
- Without an IP address, devices wouldn't know **where to send or receive data**, making internet communication impossible.
- Types of IP addresses:

Type	Description
IPv4	Older version, 32-bit, like 192.168.0.1
IPv6	Newer version, 128-bit, like 2001:0db8:85a3:0000:0000:8a2e:0370:7334

MAC ADDRESS

- A **MAC Address (Media Access Control Address)** is a **unique physical identifier assigned to a network device** by its manufacturer.
- It is stored in the **network interface card (NIC)** of the device and is used for communication within a **local network (LAN)**.
- A **MAC address** is a **permanent ID number of a device** in a network.

- Real life example:

Student : Device

Register Number : MAC address

- Even if a student changes classroom or city, their **register number remains the same**.
- Similarly, even if your device connects to another Wi-Fi or network, its **MAC address stays the same**.

DIFFERENCE BETWEEN IP AND MAC ADDRESS

Feature	IP Address	MAC Address
Meaning	Temporary address given to a device in a network	Permanent address built into the device
Changes?	Yes, it can change (like when you connect to different Wi-Fi)	No, it never changes
Who gives it?	Network (Wi-Fi/Internet provider)	Manufacturer (company that made the device)
Use	To communicate on the internet / outside network	To communicate inside local network (LAN)
Example	192.168.1.5	A4:6F:12:9B:33:E7
Type	Logical address	Physical hardware address

PORT

- A **port** is a **logical communication endpoint** that helps a computer identify which application or service should receive network data.

Simple explanation:

Think of a **house address**:

- The **IP address** tells *which house* to send the letter to.
- The **port** tells *which person/room inside the house* should receive the letter.

Without Port	With Port
Device receives data but doesn't know which app should handle it	Device sends data to correct app (browser, email, gaming, etc.)

NETWORK PROTOCOL

- A **network protocol** is a set of rules and guidelines that computers follow to **communicate over a network**.
- Just like people follow a **language** or **grammar rules** to talk to each other, devices in a network follow **protocols** to send and receive data correctly.
- Network protocols define how devices should:
 - ➔ Start a connection
 - ➔ Send data
 - ➔ Receive data
 - ➔ Detect errors
 - ➔ End communication

COMMON NETWORK PROTOCOLS

Protocol	Purpose
HTTP	Browsing websites
HTTPS	Secure website browsing
FTP	File transfer
SMTP	Sending emails
TCP	Reliable data transmission
IP	Identifying devices and addressing
DNS	Convert domain name to IP address

CONNECTION ORIENTED SERVICE

A **connection-oriented service** requires creating a **connection before sending data**.

Example : Like making a **phone call** — first you connect to the person, then you start talking.

How it works:

- ✓ Connection is established
- ✓ Data is transferred
- ✓ Connection is closed

Key features:

- ✓ communication
- ✓ Data arrives in order
- ✓ No data loss (if lost, it is retransmitted)
- ✓ Slower due to connection setup

Real protocol example : TCP

CONNECTIONLESS SERVICE

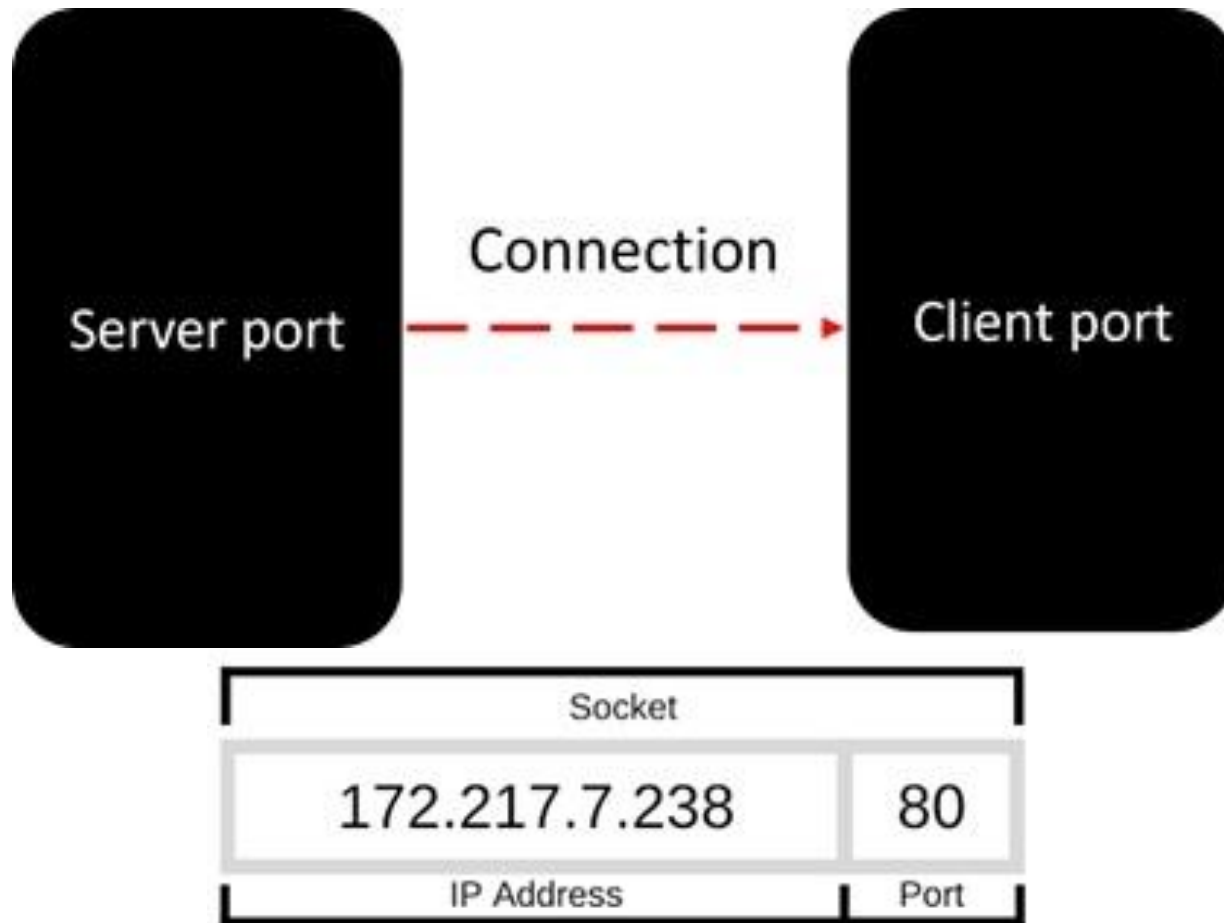
- A **connectionless service** sends data **without establishing a connection**.
- **Simple example** : Like **sending a text message or letter** — you just send it without checking if the receiver is ready.
- **How it works:**
 - ✓ Data is sent immediately — no connection setup
 - ✓ Receiver may or may not get it
- **Key features:**
 - ✓ Faster communication
 - ✓ Less reliable (data may be lost)
 - ✓ Data may arrive out of order

Real protocol example: **UDP (User Datagram Protocol)** is connectionless.

SOCKET

- A socket is the connection point that allows applications on different devices to communicate over a network.
- A socket is like a **door** in a computer through which data goes **in and out** when communicating with other devices.
- To send data over a network:
 - ✓ IP address tells **which device**
 - ✓ Port tells **which application**
 - ✓ **Socket** combines both to **create a communication link**
- A socket allows two programs to:
 - ✓ Open communication
 - ✓ Send data and Receive data

SOCKET





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